

WEST WOODWARD, OK, USA

WMO: 722152

Lat: 36.437N

Lon: 99.521W

Elev: 667

StdP: 93.57

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 53957

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-8.9	-17.8	0.8	-5.8	-16.1	1.0	-4.4	14.3	13.3	12.8	10.4	4.1	0	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	39.1	20.7	37.6	20.8	36.2	20.8	23.7	31.5	23.0	31.1	22.4	30.6	5.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	18.1	26.3	21.2	17.2	25.8	20.2	16.1	25.8	74.4	31.3	71.6	31.3	69.5	30.4	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.8	10.8	DB	-16.1	40.8	3.0	2.1	-18.2	42.2	-19.9	43.4	-21.6	44.6	-23.7	46.1	(4)
(5)			WB	-16.7	24.6	2.7	0.7	-18.7	25.1	-20.3	25.5	-21.8	25.9	-23.8	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	3.6	4.5	10.5	14.8	19.9	25.9	28.3	27.3	23.2	16.0	9.4	3.6	(6)	
	DBStd	10.28	5.81	6.58	5.96	5.57	5.03	3.43	3.18	3.55	3.96	5.49	5.58	5.44	(7)	
	HDD10.0	765	210	174	66	19	2	0	0	0	0	13	76	206	(8)	
	HDD18.3	2116	458	389	249	133	43	1	0	0	7	110	271	456	(9)	
	CDD10.0	2828	11	19	82	162	309	478	568	538	396	198	59	8	(10)	
	CDD18.3	1137	0	1	6	26	92	228	310	279	153	37	4	0	(11)	
	CDH23.3	13710	6	24	105	351	1024	2731	4047	3383	1563	416	61	1	(12)	
	CDH26.7	7091	1	6	21	128	436	1411	2336	1887	708	144	14	0	(13)	
(14) Wind	WSAvg	5.1	4.9	4.9	5.7	6.1	5.4	5.4	4.5	4.4	4.9	5.1	5.1	4.7	(14)	
(15) Precipitation	PrecAvg	615	19	20	43	51	83	96	60	69	47	54	31	27	(15)	
	PrecMax	764	71	57	150	176	187	187	217	175	107	221	108	109	(16)	
	PrecMin	437	1	0	0	0	0	5	4	15	0	0	1	2	(17)	
	PrecStd	98	17	16	38	38	45	51	42	38	28	65	32	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	26.4	29.1	33.7	37.0	40.3	42.1	41.3	37.6	32.9	28.0	21.2	(19)	
		MCWB	10.4	10.7	13.7	16.1	19.0	19.4	20.7	20.4	19.6	17.7	13.4	10.6	(20)	
	2%	DB	19.1	22.3	26.3	30.3	33.7	37.4	39.8	39.0	35.0	30.9	24.9	17.7	(21)	
		MCWB	8.2	10.1	13.6	15.1	18.5	20.1	20.6	21.1	20.0	17.2	12.4	8.9	(22)	
	5%	DB	16.2	17.9	23.0	27.4	31.4	35.2	37.6	37.4	32.7	27.7	22.0	15.1	(23)	
		MCWB	7.0	8.9	12.0	15.0	18.3	20.3	20.9	21.5	19.7	16.2	12.0	7.2	(24)	
	10%	DB	12.8	15.0	21.0	24.8	28.8	33.0	36.2	35.1	31.1	25.2	18.0	12.4	(25)	
		MCWB	5.7	7.3	11.8	14.4	17.8	20.4	21.2	21.3	19.4	15.6	10.7	6.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	14.1	17.9	19.7	22.2	23.8	25.1	24.5	23.1	21.4	16.1	12.9	(27)	
		MCDB	20.1	18.4	23.7	25.5	29.2	31.1	32.7	32.5	29.7	28.0	21.2	15.6	(28)	
	2%	WB	9.3	11.3	16.3	18.2	21.1	22.8	23.7	23.7	21.9	19.8	14.5	10.2	(29)	
		MCDB	18.3	20.0	22.3	24.6	29.1	30.7	31.1	31.8	29.1	26.0	20.7	15.5	(30)	
	5%	WB	7.6	9.7	14.7	16.9	20.2	22.2	22.9	23.0	21.2	18.5	13.1	8.2	(31)	
		MCDB	15.6	17.6	20.2	23.2	27.7	30.4	31.2	31.4	28.8	24.7	19.7	14.9	(32)	
	10%	WB	6.0	7.8	12.5	15.6	19.4	21.5	22.3	22.4	20.5	16.8	11.6	6.5	(33)	
		MCDB	12.5	14.4	20.0	22.6	26.2	30.0	30.8	30.5	28.0	23.2	17.9	12.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	13.5	14.1	13.8	12.8	12.7	12.8	12.8	12.7	13.6	13.9	12.0	(35)	
		MCDBR	17.9	19.0	17.2	17.4	15.5	14.3	14.4	15.2	14.4	16.2	17.7	16.3	(36)	
	5% WB	MCWBR	9.6	9.7	7.8	7.5	5.7	4.2	3.4	3.7	4.4	6.6	8.1	9.0	(37)	
		MCWBR	17.4	17.5	14.7	13.8	13.0	12.5	12.1	12.6	12.0	13.2	14.6	14.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.271	0.280	0.320	0.345	0.360	0.396	0.401	0.389	0.357	0.330	0.291	0.272	(40)	
		taud	2.552	2.533	2.429	2.394	2.423	2.363	2.371	2.364	2.432	2.488	2.535	2.554	(41)	
	Ebn at Noon		941	969	949	937	921	884	877	882	897	895	905	916	(42)	
		Edh at Noon	76	88	107	117	116	123	121	119	106	91	77	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.83	3.58	4.74	5.85	6.54	7.19	7.06	6.32	5.32	3.96	3.01	2.46	(44)	
	RadStd		0.26	0.44	0.38	0.39	0.59	0.53	0.36	0.34	0.29	0.52	0.34	0.22	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page